Lecture 5: Programming tools

Introduction

- A programming tool is a software utility or language feature used in C programming to assist in writing source code, compiling programs, detecting errors, and improving program quality.

Lesson objectives:

By the end of this lesson, the learner will be able to:

- Describe various programming tools.
- Identify the role of linkers, debuggers and loaders.
- Identify IDEs suitable for C language.
- Download and install necessary programming tools.

Lecture 5: Programming tools

Components used to create programs

a) **Text Editors=** Used to type and edit source code. Examples include Notepad, Notepad++, and VS Code.

b) **Language Translators=** Convert source code into machine understandable form. This includes compilers, interpreters, and assemblers.

c) **Compilers=** Translate the entire C program into machine code before execution and report syntax errors.

d) **Interpreters=** Translate and execute the program line by line, commonly used for scripting languages.

e) **Assemblers=** Convert assembly language programs into machine code.

f) **Debuggers=**Assist in detecting and correcting logical and runtime errors by allowing step by step program execution.

Lecture 5: Programming tools

Components used to create programs

g) Integrated Development Environments (IDEs)=Software applications that combine multiple tools such as editors, compilers, debuggers, and build tools in one environment. Examples include Code Blocks, Dev C++, and Visual Studio.

h) Linkers=Combine object files and link the program with required libraries and built in functions.

i) Loaders=Load the executable program into main memory and prepare it for execution.

j) Libraries= Collections of pre-written code that provide reusable functions, such as the C standard library.

k) Build Tools=Automate the compilation and linking process, especially for large programs e.g. Make.

l) Version Control Systems= Manage changes to source code over time and support collaboration e.g. Git.

Lecture 5: Programming tools

Activity 5:

Complete the table below with regard to programming tools. The first row has been completed for you.

Programming tool	Use
Linker	Links the program to built- in functions
	Translate assembly language instructions
Debuggers	
	Type and edit the source code
Interpreter	
	Load the program into the main memory
GitHub	
	Removing errors

Lecture 5: Programming tools

Exercise 5:

a) Briefly explain the role of the following programming tools.
   a) Compiler
   b) Linker
   c) Debugger

b) Differentiate an open source and proprietary programming tool. Give an example in each case.

Reference:

1. An introduction to programming by Wu Thomas, Norman and Theodore: McGrill (1997).
2. C programming language by Brian W. Kernighan, Dennis Ritchie, 2nd ed: prentice hall(2003).
3. C how to program by H.M. Deitel and P.J.Deitel, 3rd ed: Prentice hall(2001).
4. https://www.w3schools.in/c-tutorial/
5. https://www.tutorialspoint.com/cprogramming/index.htm
6. https://www.learn-c.org/
7. https://www.studysmarter.co.uk/explanations/computer-science/computer-programming/c-main/

Thank you